26900565 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCZGY302, MG5DSCZIM301 - FUNDAMENTALS OF GENETICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What phenotypic ratio is obtained when the F_1 hybrid Tt is crossed with the recessive parent tt? [A] / [CO1]

   a). All tall b). All dwarf

   c). 3 tall : 1 dwarf d). 1 tall : 1 dwarf

2. A gene in Drosophila affects wing size, eye colour, fertility and lifespan. Which genetic phenomenon does this situation demonstrate? [A] / [CO1]

   a). Polygenic inheritance b). Codominance

   c). Pleiotropism d). Linkage

3. In roan cattle, both red and white hairs are expressed. This illustrates: [U] / [CO1]

a). Incomplete dominance b). Co-dominance

c). Complementary gene action d). Supplementary gene action

4. Recombination frequency is expressed as: [K] / [CO2]

a). Parental/total offspring ratio b). Chromosomes exchanged

c). Percentage recombinant offspring d). Dominant/recessive ratio

5. A Drosophila has an X/A ratio of 1.0. Which sex type is expected? [A] / [CO2]

a). Normal male b). Normal female

c). Intersex d). Supermale

6. A frameshift mutation changes the: [K] / [CO3]

a). DNA colour b). Cell shape

c). Reading frame d). Chromosome number

7. Which symbol represents a male in a human pedigree? [K] / [CO4]

a). Circle b). Square

c). Triangle d). Oval

8. Which type of chromatin is generally less condensed? [U] / [CO4]

a). Heterochromatin b). Centromeric chromatin

c). Euchromatin d). Telomeric chromatin

9. Which of the following is commonly associated with Klinefelters syndrome? [U] / [CO5]

a). Gynaecomastia b). Achondroplasia

c). Ovarian dysgenesis d). Polydactyly

10. Cleft lip primarily involves a defect in the: [K] / [CO5]

a). Roof of the mouth b). Upper lip

c). Tongue d). Lower jaw

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. Distinguish between complementary and supplementary gene interaction. [U] / [CO1]

12. Define homozygous and heterozygous conditions with examples. [K] / [CO1]

13. What is meant by sex-influenced inheritance? How does it differ from sex-linked inheritance? [U] / [CO1]

14. Demonstrate the possible offspring genotypes produced in the cross RrYy × rryy. [U] / [CO1]

15. Describe the sexual dimorphism in *Bonellia viridis.* [K] / [CO2]

16. Explain why a freemartin is usually sterile. [U] / [CO2]

17. Identify and explain the significance of a darkly stained chromatin body in the nucleus of a female somatic cell. [A] / [CO2]

18. Explain why an A→G substitution is classified as a transition, whereas an A→T substitution is classified as a transversion. [U] / [CO3]

19. Describe how a mutation occurring in a somatic cell can become restricted to a particular tissue or cell lineage. [U] / [CO3]

20. Define mutagens and list their major types with one suitable example each. [K] / [CO3]

21. List the major levels of chromosome organization. [K] / [CO4]

22. Explain the significance of sequencing the human genome. [U] / [CO5]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. A woman with blood group A gives birth to a child with blood group AB. Two men are considered possible fathers: one has blood group B and the other has blood group O. Which man could genetically be the father? Justify your answer. [A] / [CO1]

24. List the seven pairs of contrasting characters studied by Mendel in pea plants, indicating the dominant and recessive trait in each pair. [K] / [CO1]

25. Describe Morgan's experiments in Drosophila involving grey-bodied, long-winged and black-bodied, vestigial-winged flies, with a neat labelled illustration, and explain his findings. [U] / [CO2]

26. Describe chromosomal sex determination in humans, indicating the sex-chromosome constitution and gamete types of males and females. [U] / [CO2]

27. List the major numerical chromosomal aberrations and state their basic characteristics. [K] / [CO3]

28. With the help of a labelled diagram, describe the structure of a lampbrush chromosome. [U] / [CO4]

29. Compare phenylketonuria, alkaptonuria and albinism with respect to the defective enzyme or biochemical process, metabolic consequence and major clinical manifestations. [U] / [CO5]

30. Briefly explain the genetic and physiological basis of cystic fibrosis. [U] / [CO5]

[5x6 = 30]